

CS2601 Linear and Convex Optimization

Homework 2 Solution

Due: 2021.10.9

1. Let $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ be an affine function from $\mathbb{R}^n$ to $\mathbb{R}^m$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Show that if $C \subset \mathbb{R}^m$ is convex, so is its inverse image $f^{-1}(C) \triangleq \{\mathbf{x} : f(\mathbf{x}) \in C\}$.

Proof. Let $\mathbf{x}_1, \mathbf{x}_2 \in f^{-1}(C)$ and $\theta \in [0, 1]$. By definition, $f(\mathbf{x}_i) \in C$. By convexity of C ,

$$f(\theta\mathbf{x}_1 + \bar{\theta}\mathbf{x}_2) = \mathbf{A}(\theta\mathbf{x}_1 + \bar{\theta}\mathbf{x}_2) + \mathbf{b} = \theta(\mathbf{A}\mathbf{x}_1 + \mathbf{b}) + \bar{\theta}(\mathbf{A}\mathbf{x}_2 + \mathbf{b}) = \theta f(\mathbf{x}_1) + \bar{\theta} f(\mathbf{x}_2) \in C$$

Thus

$$\theta\mathbf{x}_1 + \bar{\theta}\mathbf{x}_2 \in f^{-1}(C)$$

and $f^{-1}(C)$ is convex. □

2. Suppose C_1, C_2 are two nonempty convex sets with $C_1 \cap C_2 = \emptyset$. Show $C = C_1 - C_2 = \{\mathbf{x}_1 - \mathbf{x}_2 : \mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2\}$ is a nonempty convex set and $\mathbf{0} \notin C$.

Proof. Since $C_1, C_2 \neq \emptyset$, there exists $\mathbf{x}_i \in C_i$, $i = 1, 2$. Then $\mathbf{x}_1 - \mathbf{x}_2 \in C$, so $C \neq \emptyset$.

If $\mathbf{0} \in C$, then $\mathbf{0} = \mathbf{x}_1 - \mathbf{x}_2$ for $\mathbf{x}_i \in C_i$, which implies $\mathbf{x}_1 = \mathbf{x}_2 \in C_1 \cap C_2$, contradicting the assumption $C_1 \cap C_2 = \emptyset$. So $\mathbf{0} \notin C$.

Suppose $\mathbf{y}_1 = \mathbf{x}_{1,1} - \mathbf{x}_{1,2}, \mathbf{y}_2 = \mathbf{x}_{2,1} - \mathbf{x}_{2,2} \in C$, where $\mathbf{x}_{i,j} \in C_j$, $i, j \in \{1, 2\}$. By convexity of C_j ,

$$\theta\mathbf{x}_{1,j} + \bar{\theta}\mathbf{x}_{2,j} \in C_j.$$

Thus

$$\theta\mathbf{y}_1 + \bar{\theta}\mathbf{y}_2 = (\theta\mathbf{x}_{1,1} + \bar{\theta}\mathbf{x}_{2,1}) - (\theta\mathbf{x}_{1,2} + \bar{\theta}\mathbf{x}_{2,2}) \in C,$$

and hence C is convex. □

3. Suppose C is a convex set.

- (a). Show that its interior $\text{int } C$ is convex.
- (b). Show that its closure $\bar{C}$ is convex.

Proof. (a). Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \text{int } C$. Then there exists $r > 0$ such that $B(\mathbf{x}_i, r) \subset C$, $i = 1, 2$. Consider $\mathbf{x}_\theta = \theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2$ for $\theta \in [0, 1]$. Now we show $B(\mathbf{x}_\theta, r) \subset C$ (Try to visualize it in 2D to get some intuition). Note

$$B(\mathbf{x}_\theta, r) = \{\mathbf{x}_\theta + r\mathbf{u} : \mathbf{u} \in B(\mathbf{0}, 1)\}$$

Since $\mathbf{x}_i + r\mathbf{u} \in B(\mathbf{x}_i, r) \subset C$, $i = 1, 2$, and C is convex,

$$\mathbf{x}_\theta + r\mathbf{u} = \theta(\mathbf{x}_1 + r\mathbf{u}) + \bar{\theta}(\mathbf{x}_2 + r\mathbf{u}) \in C$$

Therefore, $B(\mathbf{x}_\theta, r) \subset C$ and $\mathbf{x}_\theta \in \text{int } C$.

(b). Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \bar{C}$. There exists a sequence $\{\mathbf{x}_{i,k}\}_{k \geq 1}$ such that $\mathbf{x}_{i,k} \rightarrow \mathbf{x}_i$ as $k \rightarrow \infty$, $i = 1, 2$. For $\theta \in [0, 1]$, by the convexity of C ,

$$\theta \mathbf{x}_{1,k} + \bar{\theta} \mathbf{x}_{2,k} \in C.$$

Let $k \rightarrow \infty$,

$$\theta \mathbf{x}_{1,k} + \bar{\theta} \mathbf{x}_{2,k} \rightarrow \theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2$$

so $\theta \mathbf{x}_1 + \bar{\theta} \mathbf{x}_2 \rightarrow \bar{C}$ and hence $\bar{C}$ is convex. □

4. Prove the convex hull $\text{conv } S$ of S is the set of all convex combinations of points in S by completing the following steps.

(a). Let

$$C = \left\{ \sum_{i=1}^m \theta_i \mathbf{x}_i : m \in \mathbb{N}; \mathbf{x}_i \in S, \theta_i \geq 0, i = 1, \dots, m; \sum_{i=1}^m \theta_i = 1 \right\}$$

Show that C is convex.

(b). Show that $C \subset \text{conv } S$ and conclude $C = \text{conv } S$.

Proof. (a). Let $\mathbf{y}_1, \mathbf{y}_2 \in C$ and $\alpha \in [0, 1]$. Then

$$\mathbf{y}_1 = \sum_{i=1}^{m_1} \theta_{1,i} \mathbf{x}_{1,i}, \quad \mathbf{y}_2 = \sum_{i=1}^{m_2} \theta_{2,i} \mathbf{x}_{2,i}$$

where $\mathbf{x}_{i,j} \in S$, $\theta_{i,j} \geq 0$, $i = 1, 2, j = 1, 2, \dots, m_i$ and $\sum_{j=1}^{m_i} \theta_{i,j} = 1$, $i = 1, 2$. Consider

$$\alpha \mathbf{y}_1 + \bar{\alpha} \mathbf{y}_2 = \sum_{i=1}^{m_1} \alpha \theta_{1,i} \mathbf{x}_{1,i} + \sum_{i=1}^{m_2} \bar{\alpha} \theta_{2,i} \mathbf{x}_{2,i}.$$

Since $\alpha \theta_{1,i} \geq 0$, $\bar{\alpha} \theta_{2,i} \geq 0$, and

$$\sum_{i=1}^{m_1} \alpha \theta_{1,i} + \sum_{i=1}^{m_2} \bar{\alpha} \theta_{2,i} = \alpha + \bar{\alpha} = 1,$$

$\alpha \mathbf{y}_1 + \bar{\alpha} \mathbf{y}_2$ is a convex combination of $\mathbf{x}_{1,1}, \dots, \mathbf{x}_{1,m_1}, \mathbf{x}_{2,1}, \dots, \mathbf{x}_{2,m_2}$ and hence belongs to C . Therefore C is convex.

(b). Since $\text{conv } S$ is convex, it contains all convex combinations of points in S , so $C \subset \text{conv } S$. Since C is convex, $C = \text{conv } S$ by the minimality of $\text{conv } S$. □

5. Let $\mathbf{x}_0, \dots, \mathbf{x}_K$ be distinct points in $\mathbb{R}^n$. Let V be the set of points that are closer in Euclidean distance to $\mathbf{x}_0$ than $\mathbf{x}_1, \dots, \mathbf{x}_K$, called the **Voronoi region** around $\mathbf{x}_0$ with respect to $\mathbf{x}_1, \dots, \mathbf{x}_K$, i.e.

$$V = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2, i = 1, 2, \dots, K\}.$$

Show that V is a polyhedron by identifying $\mathbf{A}$ and $\mathbf{b}$ such that $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$. (You don't have to draw it in the submission, but try to visualize V for $\mathbb{R}^2$.)

Proof. Note that

$$\begin{aligned} \|\mathbf{x} - \mathbf{x}_0\|_2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2 &\iff \|\mathbf{x} - \mathbf{x}_0\|_2^2 \leq \|\mathbf{x} - \mathbf{x}_i\|_2^2 \\ &\iff (\mathbf{x} - \mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) \leq (\mathbf{x} - \mathbf{x}_i)^T(\mathbf{x} - \mathbf{x}_i) \\ &\iff -2\mathbf{x}_0^T\mathbf{x} + \mathbf{x}_0^T\mathbf{x}_0 \leq -2\mathbf{x}_i^T\mathbf{x} + \mathbf{x}_i^T\mathbf{x}_i \\ &\iff 2(\mathbf{x}_i - \mathbf{x}_0)^T\mathbf{x} \leq \mathbf{x}_i^T\mathbf{x}_i - \mathbf{x}_0^T\mathbf{x}_0 \end{aligned}$$

Let

$$\mathbf{A} = \begin{pmatrix} 2(\mathbf{x}_1 - \mathbf{x}_0)^T \\ \vdots \\ 2(\mathbf{x}_K - \mathbf{x}_0)^T \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \mathbf{x}_1^T\mathbf{x}_1 - \mathbf{x}_0^T\mathbf{x}_0 \\ \vdots \\ \mathbf{x}_K^T\mathbf{x}_K - \mathbf{x}_0^T\mathbf{x}_0 \end{pmatrix}$$

Then $V = \{\mathbf{x} : \mathbf{A}\mathbf{x} \leq \mathbf{b}\}$, so V is a polyhedron. □